

BUILDER JOURNAL

Architecture & Pipeline Documentation

FELLOW

M. Abdullah Nazir

TRACK & ASSIGNMENT

Computer Vision · Assignment 8

PROJECT FOCUS

Bottle Detection System

SYSTEM OVERVIEW

Project Description: An engineering-focused Bottle Detection System powered by a lightweight 7-stage architectural pipeline designed for high-precision, real-time edge processing applications.

Image → Preprocessing → YOLO → Boxes → Filter → Visualization → Output

CORE ARCHITECTURE SPECS

PROPERTY	DETAIL
Model Configuration	YOLOv8 Nano (YOLOv8n) — 3.0M params 8.1 GFLOPs
Framework Stack	Python · Ultralytics · Streamlit · OpenCV · PyTorch · NumPy · Pillow
Pipeline Execution	7 Distinct Sequential Stages
Model Performance	mAP50: 87.1% Precision: 93.5% Recall: 79.2%
Inference Speed	9.3ms per image (Real-time capable on Quadro T1000 GPU)
Deployment Mode	Streamlit Web Application (In-memory, Stateless execution)

01/ PIPELINE CORE

Every image or frame uploaded by the user passes through each stage in sequence from top to bottom. Below is the detailed breakdown of the 7-stage pipeline:

01 / INPUT

IMAGE INPUT

User uploads image, video frame, or webcam capture. Decoded via `cv2.imdecode()` into a uint8 BGR matrix of shape (H, W, 3).

02 / PREPROCESS

PREPROCESSING

Resizes to 640×640 via letterboxing to preserve aspect ratio. Normalizes pixels to 0–1, reorders channels to RGB, and batches into an NCHW PyTorch tensor.

03 / INFERENCE

YOLO MODEL

Executes forward pass through the YOLOv8n network (Backbone, Neck, and Anchor-free decoupled head) within 9.3ms to produce thousands of raw candidate boxes.

04 / DECODE

BBOX DECODING

Converts bounding coordinates from center-based format to absolute pixels (x1, y1, x2, y2). Applies Non-Maximum Suppression (NMS) with an IoU threshold of 0.45 to eliminate duplicates.

05 / FILTER

CONFIDENCE FILTER

Applies an empirically tuned confidence threshold of 0.40 (balancing 93.4% Precision and 81.1% Recall) to filter out weak predictions. User adjustable from 0.10 to 0.95.

06 / VISUALIZE

VISUALIZATION

Draws bounding boxes and confidence labels text over the original matrix using `cv2.rectangle()` and `cv2.putText()`. Converts BGR back to RGB for display.

07 / OUTPUT

OUTPUT GENERATION

Renders the side-by-side comparison in the Streamlit UI dashboard with detection tags and a historical breakdown panel. Supports direct in-memory lossless PNG/MP4 binary downloads via BytesIO.

02/ DATA FLOW ARCHITECTURE

STAGE	INPUT MATRIX / SOURCE	OUTPUT TARGET	FORMAT & PRECISION
01 Input	File / Camera Stream	Raw BGR Image Array	uint8 (H, W, 3)
02 Preprocess	Raw BGR Array	Model Tensor Box	float32 (1, 3, 640, 640) 0-1
03 YOLO Inference	Model Tensor Box	Raw Candidate Boxes	Thousands of bounding vectors
04 BBox Decode	Raw Predictions	Decoded Uniques	[x1, y1, x2, y2, conf, cls] list
05 Conf Filter	All Decoded Unique Boxes	Final Confident Detections	Boxes with confidence ≥ 0.40
06 Visualize	Original Image + Filtered Boxes	Annotated Composite Image	RGB Image with drawn boxes
07 Output	Annotated Image Composite	Browser Display + Download	Lossless PNG File / Streamlit UI

03/ ENGINEERING DESIGN DECISIONS

1. Stateless Pipeline Design

Every pipeline operation reads exclusively from the raw uploaded source bytes rather than mutating states across historical blocks. This eliminates quality degradation over chained adjustments and allows users to change filters interactively with instantaneous zero-loss feedback loops.

2. In-Memory Processing Framework

No data is written to or read from physical storage disks during the execution run. All image buffers are processed natively in RAM using high-performance NumPy vectors and binary BytesIO memory blocks, increasing throughput and ensuring cloud compatibility.

3. Optimized Transfer Learning Sequence

The network was initialized using pre-trained weights from the COCO dataset (80 classes, 330k images) to inherit high-level primitive spatial features like edge contours and surface gradients. Fine-tuning the decoupled head on 2,037 custom bottle images isolated training down to just 59 minutes on a Quadro T1000 GPU.

4. Anchor-Free Head Decoupling

YOLOv8 removes legacy anchor boxes in favor of direct continuous coordinate regression points. This eliminates manual aspect ratio hyperparameter optimization cycles and vastly improves precision on overlapping or abnormally shaped bottle structures.